



**SUNYANI TECHNICAL UNIVERSITY(STU)
FACULTY OF APPLIED SCIENCE & TECHNOLOGY**

DEPARTMENT OF COMPUTER SCIENCE

DRAFT OUTLINE FOR STUDENTS PROJECT WORK

JULY, 2023

SUNYANI TECHNICAL UNIVERSITY
FACULTY OF APPLIED SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE

**CABBAGE CROP DISEASE DETECTION MODEL USING
CONVOLUTIONAL NEURAL NETWORK**

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PROJECT TOPIC:
CABBAGE CROP DISEASE DETECTION MODEL USING
CONVOLUTIONAL NEURAL NETWORK

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PROJECT SUBMITTED TO THE DEPARTMENT OF COMPUTER
SCIENCE, FACULTY OF APPLIED SCIENCE AND TECHNOLOGY IN
PARTIAL FULFILLMENT FOR THE REQUIREMENT OF THE
AWARD OF NATIONAL HIGHER DIPLOMA IN COMPUTER
SCIENCE

JULY, 2023

DECLARATION

We hereby declare that, except for the reference to other people's work, which has been duly acknowledged, this research work consists of our work produced from research undertaken under supervision and that no part nor full, has been published or presented for any degree elsewhere

NAME OF STUDENT	INDEX NUMBER	SIGNATURE	DATE
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CERTIFICATION

This is to certify that the project report titled “ CABBAGE CROP DISEASE DETECTION MODEL USING CONVOLUTIONAL NEURAL NETWORK” is an authentic record and work done by Jenifer Apietu, Narh Frederick and Saviour Soku, and submitted in partial fulfilment of the requirement for the Higher National Diploma / Bachelor of Technology in Computer Science / Information and Communication Technology at Sunyani Technical University (STU).

MRS. AWUTI MENSAH
(SUPERVISOR)

DATE

DR. BEN BEKLISI KWAME AYAWLI
(HEAD OF DEPARTMENT)

DATE

DEDICATION (OPTIONAL)

The project work can be dedicated to anybody at all according to the choice of the candidate.

ACKNOWLEDGEMENT

The student is expected to thank or acknowledge individuals, people and institutions who helped to make the project a success.

ABSTRACT

The abstract is more or less the summary of the project. It highlights the most important aspects within the project work such as, but not limited to the following: introduction, specific objectives, methodology and results. An abstract is a concise summary of a research paper or entire thesis. It is an original work, not an excerpted passage. An abstract must be fully self-contained and make sense by itself, without further reference to outside sources or to the actual paper. It highlights key content areas, your research purpose, the relevance or importance of your work, and the main outcomes. It is a well-developed single paragraph of approximately 250 words in length, which is indented and single spaced. The function of the abstract is to outline briefly all parts of the paper. Although it is placed at the beginning of your paper, immediately following the title page, the abstract should be the last thing that you write, once you are sure of the conclusions you will reach.

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CHAPTER ONE

INTRODUCTION

(Brief introduction to indicate what this chapter presents)

1.1 Background of the Study/ Project

The background typically includes a comprehensive overview of the subject matter, highlights of the existing knowledge, gaps, and the rationale for conducting the research. The background section aims to demonstrate the need for the project, establish its relevance, and outline its significance in the broader field of study.

1.2 Statement of Problem

This section should provide a concise and clear description of the specific issue, concerns or gap in knowledge that the study aims to address.

1.3 Objectives of the Study

1.3.1 General Objective

Present the broad purpose of the research and provides a clear direction for the entire project.

1.3.2 Specific Objectives

- Specify the basic objectives to be addressed by your research/project.
- Your specific objectives must be SMART (Specific, Measurable, Achievable, Relevant and Time bound)

1.4 Scope of the Project

- Explain the extent, breadth, or coverage of the study. It defines the boundaries of the research in terms of the topics, variables, and parameters that will be included in the investigation.
- Present the aspects of the problem that have been examined to define the overall context for the study. The scope should address questions such as:
 - What topics or variables are investigated?
 - What specific research questions or objectives are addressed?
 - What time period, geographical area, or population is considered?

- What theories, concepts, or frameworks are applied?

1.5 Significance of the Project

Discuss the importance, relevance, and potential impact of the study. Highlight why the research is valuable and how it contributes to the field of study, existing knowledge, practical applications, or the broader community.

1.6 Limitations and Delimitations

1.6.1 Limitations

Discuss the constraints, weaknesses, or factors that may impact the scope, implementation, or generalizability of the study.

1.6.2 Delimitations

Present exclusions, limitations, or specific boundaries set for the project. Define what aspects or factors are intentionally excluded or restricted from the study.

1.7 Organization of Work

Briefly explain the content of each chapter and how they are organized in the project report.

CHAPTER TWO

LITERATURE REVIEW

(Brief introduction to indicate what this chapter presents)

It should be noted that this chapter has no exact sub headings. Candidates determine the sub headings of this chapter based on the topic, problem statement, and the objectives of the research/Project. However, the candidates are expected to cover the following;

- To review the relevant state-of-the-art literature on the chosen topic and associated technologies.
- To review and identify gaps in related works
- To relate what they are proposing to do, to what others have done to date
- To make a clear case of the existence of the research gap indicated in the problem statement.

CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN

(Chapter's main heading for those who strictly designed software solutions)

Brief Introduction to indicate what this chapter presents

3.1 System Analysis

3.1.1 Tools used for the System Analysis

- Indicate the tools used for your data collection and analysis (eg. Review of documentation, observation). Conducting interviews, questionnaires etc
- Why did you use the selected tool(s) for the data
- Describe how you used the tools for the data collection for the analysis of the existing systems.

3.1.2 Description of the Existing System

- Describe the existing system by indicating clearly the challenges and deficiency of the existing system. This should be inline with your problem statement

3.1.3 Problem Definition

- Clearly define the specific problem you are trying to address

3.1.4 Hardware requirement specification

3.1.5 Software requirement specification

3.1.6 Functional requirements specification

3.1.7 Non Functional requirements specification

3.1.8 Technical requirements specification

3.1.9 Feasibility analysis of the proposed system

- Economic, Technical, Behavioral and legal feasibility analysis

3.2 System Analysis

- Brief introduction
- Description of the design model chosen, reason for the choice and how you use it in your design
- Brief introduction to the design specifications

3.2.1 Input and Output Design Specification

- Indicate clearly the models of the system.
- Provide algorithms/Pseudocodes where necessary
- Indicate clearly the system design specifications using data flow diagrams (Context data flow diagram, Level 0, Level 1 and Level 2 data flow diagrams) for the proposed system, activity diagrams, use case diagrams, flow charts, decision trees etc.

3.2.3 Database Design

- Indicate your database design by specifying and describing the entities, attributes and relationship between the entities.
- Provide the database schema
- Provide a detailed entity relationship diagram

3.2.4 Data Element Dictionary

- Clearly Specify the data elements used with their data types and descriptions

NB: Candidates are not to present lecture/notes on the various software development methodologies but the specific way that they are providing a solution to their chosen problem.

CHAPTER THREE

METHODOLOGY

(Main heading for those conducting scientific research in ICT/Computer Science)

The methodology is to tell the reader exactly how you designed your study and just as importantly, why you did it this way. Importantly, this chapter should comprehensively describe and justify all the methodological choices you made in your study.

The sections of this chapter should be based on how you address the gap you identified in research. This chapter may contain the following.

- Introduction to indicate what this chapter is systematically presented.
- Provide a title indicating the framework of the chosen methodology.
- Demonstrate diagrammatically a summary of your methodology using data flow diagram or flow chart.
- Obtain headings from the diagram and provide the details by describing how each of the sections of the proposed method is designed and implemented to achieve the objectives of the research.
- Indicate all the necessary algorithms, equations, figures and tables of your methodology. You may have an algorithm representing your choice of method as well as the use of schematic diagrams.

NB: Candidates are not to present lecture/notes on the various software development methodologies but the specific way that they are providing a solution to their chosen problem.

CHAPTER FOUR

RESULTS AND DISCUSSION

Chapter 4 title for those conducting scientific research

- Brief introduction
- Concerned with assessing the effectiveness and efficiency of your solution or approach.
- Provide introduction to alert the reader what he/she expects in this chapter

4.1 Results

- Indicate the platform used in evaluating your proposed method.
- What data did you use and how did you get them?
(Note. The data used determine the reliability of your findings)
- Present your results using tables, charts and figures with detailed description
- Ensure your results address all the specific objectives outlined in chapter 1

4.2 Discussion

- Compare and discuss your results
- Indicate clearly how your method is better than others using percentages and specific values for the comparison.
- Charts, tables, and figures could be used

CHAPTER FOUR

SYSTEM DEVELOPMENT, TESTING AND DEPLOYMENT

(Title for those developing management software)

(This chapter is concerned with assessing the effectiveness and efficiency of your solution or approach. Presentation of key screen shots emanating from the testing and evaluation phase. It may also include data, measurements, observations and any other relevant results obtained during the project. The results may be presented in tables, graphs, or figures to make them more visually understandable)

Provide an introduction indicating what the reader should expect in this chapter.

4.1 System Development

- Provide brief introduction for this section
- Provide in detail the development tools used, why you choose them, and how you use them.
- Clearly specify and discuss the security mechanisms implemented in your system. Why did you choose a given security mechanism and how did you implement them.

4.2 System Testing

- Systematically discuss the testing technique used, why you use them and how you conducted the test.
- Provide the detail login credentials used for the testing.
- Provide detailed testing results for each module of the system.
- Screenshots as figures could be used to demonstrate the results interfaces of the testing. Tables could also be use.
- Note to explain the information in the figures and table
- Ensure your testing results address all the stated specific objectives in chapter 1

4.3 System Deployment

- Indicate and discuss the system deployment architecture for your system. Demonstrate this using a deployment diagram(s).
- Indicate and discuss the environment and tools required for the deployment of the system.
- Indicate the deployment type (eg. Parallel pilot, etc) for your system. Describe how it would be used for deploying your system. Indicate reason for your choice

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

Brief Introduction to inform the reader what is expected in this chapter.

5.1 Summary

- Give summary of your research project

5.2 Conclusion

- Present conclusions for your research project
- Your conclusion should be directly related to your the objectives and results/findings.
- Write your conclusion using paragraphs for each objective

5.3 Recommendation(s)

- Recommendation for future work.

REFERENCES

- ❖ The APA style shall be used

Examples

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- Jacobs, G. M., Teh, J., & Spencer, L. (2019). A proposal for facilitating more cooperation in competitive sports (ED573929). ERIC. <https://files.eric.ed.gov/fulltext/ED573929.pdf>
- News From Science. (2019, June 21). Are you a fan of astronomy? Enjoy reading about what scientists have discovered in our solar system—and beyond? This [Image attached] [Statusupdate]. Facebook. <https://www.facebook.com/ScienceNOW/photos/a.117532185107/10156268057260108/?type=3&theater>
- Svendsen, S., & Løber, L. (2020). *The big picture/Academic writing: The one-hour guide* (3rd digital ed.). Hans Reitzel Forlag. <https://thebigpicture-academicwriting.digi.hansreitzel.dk/>

APPENDICES

- ❖ Key sample codes, source documents and some diagrams

FORMATTING REQUIREMENTS

- **Margins:** 1.5 inches of left margin, 1 inch for top, bottom and right margins
- **Line spacing:** Double
- **Font:** Times New Roman
- **Font Size:** 12 for all pages except cover page which should be 14
- All headings and sub headings must be formatted **BOLD**
- Page numbering must start at 1 from chapter one (introduction) and ‘i’ from declaration to the abstract.
- Entire project work must not be less than 40 pages.
- Title page must not have page number
- From Declaration page to abstract page numbering must be in roman numerals starting from ‘i’
- Page numbering from Chapter one to the last page must be in numbers

MAIN WORK

The main work begins with the table of contents followed by the abstract.